

Vector Data Structure Implementation Problem

Description:

You are required to implement a dynamic array (Vector) that can grow automatically. The initial capacity of

Functions to implement:

- `push_back(val)`: Adds an element to the end. If the size equals capacity, allocate a new array with double
- `pop_back()`: Removes the last element.
- `get(index)`: Returns the value at the given index.
- `get_size()`: Returns the current number of elements.
- `get_capacity()`: Returns the current capacity.

Constraints:

- You cannot use built-in dynamic array libraries.
- The time complexity of `push_back` should be $O(1)$ amortized.